DESCRIPTION OF COURSEWORK

Course Code	EEE302
Course Name	Electromagnetic Field Theory and Waves
Lecturer	Associate Prof. Dr. Zulkifly Abbas
Academic Session	2023/09
Assessment Title	VSWR Measurement Technique

A. Introduction/ Situation/ Background Information

VSWR (voltage standing wave ratio) is a measure of the impedance matching between transmission lines and microwave components/device/antenna. It indicates the amount of power that can be safely delivered to microwave components without damaging it.

A low VSWR value means a better impedance match, and therefore more power being transferred efficiently between the transmission lines and microwave components. A higher VSWR value means a poorer impedance match, and therefore less power being transferred between them.

As example consider an antenna system. The VSWR measures how much energy is reflected back to the transmitter when a signal is transmitted through an antenna. A high VSWR indicates that there are problems with the antenna and/or transmission line, as a large amount of energy is being reflected back instead of being radiated by the antenna.

In a typical system, you want to minimize the VSWR to improve efficiency and ensure stability of the system.

Assignments for this coursework:

VSWR Measurement Technique(lab report)

In this experiment, the measurement of maximum and minimum voltage in standing wave shall be accomplished using VSWR meter. The measurement shall be conducted using Microwave Lab

Training mod. MW-B-C/EV. This assignment is designed to provide basic microwave measurement skills including waveguide/antenna connections, determination of the maximum and minimum electric field strengths using slotted line, determination of the VSWR of microwave standards(open, short, load). The students will also learn the basics of signal generator and receiver as well as the function of microwave diode detector and VSWR meter.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Measure/determine the characteristic of transmission lines using instruments and software or other specified tools

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All the assessment submitted must be the outcome of the student. Any form of academic misconduct is a serious offense which will be penalised by being given a zero mark for the entire assessment in question or part of the assessment in question. If there is more than one guilty party as in the case of collusion, both you and your collusion partner(s) will be subjected to the same penalty.

D. Instruction to Students

The report must be submitted (Moodle) before 31 December 2023. The report must include abstract ,introduction, theory, methodology, results &discussion, conclusion and references.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Lab Report– VSWR Measurement Technique	20%
	TOTAL	

F. Tasks

VSWR Measurement

The location of the maximum and minimum voltages of fifferent types of RF components shall be measured including terminated line, slotted line mod.MW-5, short circuit mod, MW-10, and antenna mod.MW-15 and MW-16. The value of VSWR will be calculated from the measured distances of the maximum and minimum voltages of the standing waves.

G. COMPLEX ENGINEERING PROBLEM

Description of Complex Engineering Problem	
Project Theme: VSWR Measurement Technique Assessment: Presentation/Report/Demonstration	
Attributes	**Remarks**
WP1 Depth of knowledge required Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamental-based, first principles analytical approach.	Requires students to solve impedance matching problem of antennas and waveguides which cannot be resolved without knowledge of wave propagation theoretical formulation of the VSWR and slotted line measurement technique(WK6)

WP2 Range of conflicting requirements Involve wide-ranging or conflicting technical, engineering and other issues.	Students will need to identify the locations of the maxima and minima of the slotted line based on the measured voltages
WP3 Depth of analysis required Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models	VSWR measurements on various components: slotted line mod.MW-5, short circuit mod, MW-10, and antenna mod.MW-15 and MW-16. Different components will give different locations of the maxima and minima and thus different VSWR.
WP7 Interdependence Are high level problems including many component parts or sub-problems	The VSWR technique is a high level problem which have different components: the theory of wave propagation, calculation of VSWR, the waveguide slotted line measurement, the skills required for the measurement setup, procedure and analysis.

No.	EAC PO Attribute	Description	Bloom's Taxonomy Domains	WK	WP/ EA
PLO5 / CLO3	**Modern Tool Usage**	Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities, with an understanding of the limitations	Psychomotor	WK6	WP

APPENDIX 1

MARKING RUBRICS

	Excellent (90-100%)	Good (80-89%)	Adequate (60-79%)	Poor (20-60%)	Lacking 20-0%
Abstract/ Introduction/ Theory	1.Includes complete well-written information on the background to understand the subject of the experiment and its significance. 2.Summarizes what is already known on the subject. 3. Includes the question to be answered by the lab and states hypothesis that is based on research and/or sound reasoning, described in the background. 4. Includes the description on how the hypothesis is going to be tested. 5. Deep understanding of the subject as well as creativity is demonstrated.	1.Includes strong introductory information, most major points provided. 2.Most important details in the summary provided. 3. Most important details of the hypothesis covered. 4. An idea on how to test the hypothesis is included. 5. Good understanding is demonstrated.	1.Includes good introductory information, missing some points in the summary. 2.Hypothesis is missing some minor details. 3.No description on how the hypothesis is going to be tested is provided. 4. Appropriate understanding of the subject is demonstrated.	1. Introductory information provided, missing some major points. 2. Important details of the hypothesis are missing. 3. No description on how the hypothesis is going to be tested provided. 4. Limited understanding of the subject is demonstrated.	1. Part is missing or plagiarized. OR 2.Little background information is present. 3.Incorrect information is provided. 4. For the hypothesis, important experimental details are missing, of the hypothesis is missing. 5. Poor understanding or no understanding of the subject is demonstrated.
methodology	Description (not a copy of a protocol!) is included, detailed enough so it could be repeated by another scientist. Description is not overloaded with unnecessary details.	Description included, some steps are vague or unclear OR instead of description, a step-by–step protocol is copied from manual.	The description gives generalities, enough for reader to understand how the experiment was conducted	Would be impossible to repeat the experiment, reader must guess at how the data was gathered or experiment conducted	Part is missing or plagiarized

Results	Results and data are clearly recorded, so it is easy for the reader to see trends. Excellent graphs, tables or figures with appropriate names and labels, excellent information and understanding of results, excellent interpretation of data and trends while referring to figures and tables.	Results are clear, but trends are not obvious or there are minor errors in organization. Good figures, graphs, or tables, good information results, good understanding of results, but some figure/table names or labels are missing.	Results are unclear, trends are not obvious, disorganized, but there is enough data to show the experiment was conducted. Appropriate figures, graphs or tables or drawings, some labels and figure names are missing. Only some understanding of results is evident.	Results are disorganized or poorly recorded, do not make sense; not enough data was taken to justify results. Missing or incorrect information.	Part is missing or plagiarized. Misleading, confusing and/or incorrect information. Poorly presented results.
Discussion/ Conclusions	1.Summarizes data used to draw conclusions 2.Conclusions follow data (not wild guesses or leaps of logic) 3. Hypothesis is rejected or accepted based on the data. 4. Discussion of applications or realworld connections is present. 5. Deep understanding of the subject as well as creativity is demonstrated.	Good conclusion, important information provided, or one point from Excellent is missing	Appropriate conclusion. Two points from Excellent are missing.	Limited conclusion. Important information is missing.	Part is missing or plagiarized. OR No conclusion is made. Missing important information. Little effort or reflection is shown.
References	All references and citations are correctly written and present. More than two resources are cited. Student accurately cites to the sources throughout the text.	One reference or citation is missing or incorrectly present.	Appropriate citing and reference list.	References and citations are missing or errors detract significantly from paper.	Part is missing or plagiarized. Incorrect citing.